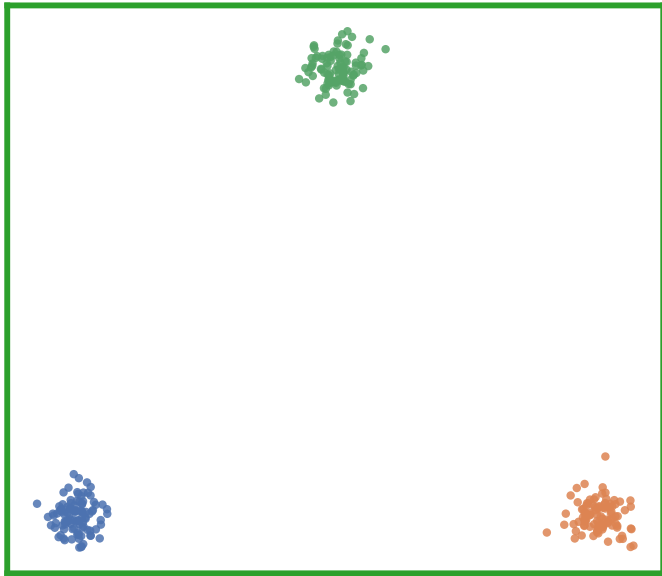
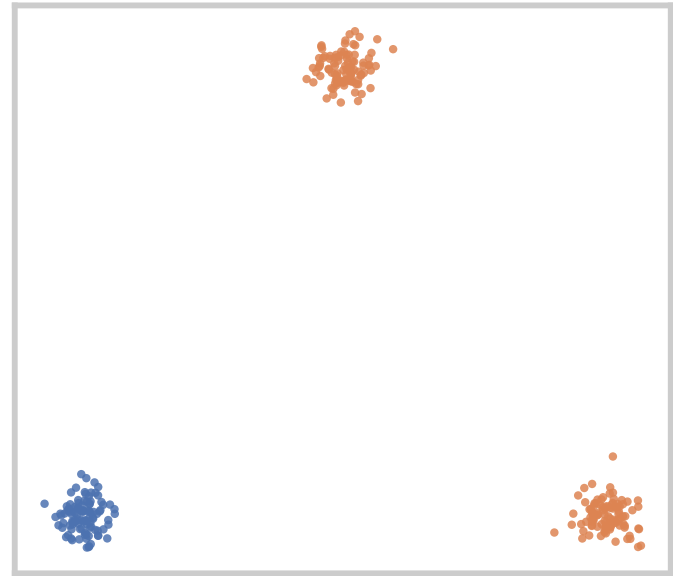


Constructive illustration of a CH failure mode

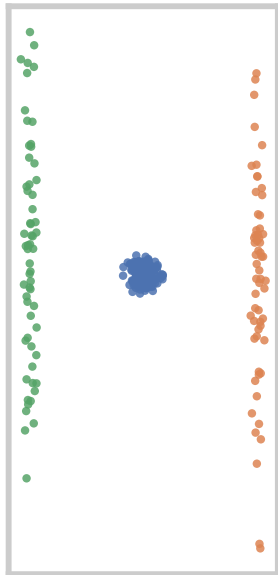
$\mathcal{F}_{\text{sph}}$: ground truth (3 clusters)
CH = 29,450 CH selects this ✓



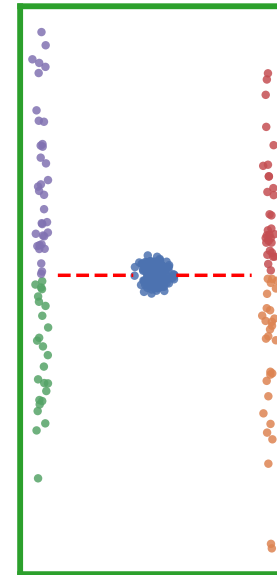
$\mathcal{F}_{\text{sph}}$: merged (2 clusters)
CH = 298 ✗



$\mathcal{F}_{\text{ell}}$: ground truth (3 clusters, AMI=1.0)
CH = 190 ✓



$\mathcal{F}_{\text{ell}}$: split (4 clusters)
CH = 373 CH selects this ✗



Top row: CH correctly prefers the ground truth. Bottom row: CH prefers the variance-reducing split (373 > 190), disagreeing with the ground truth.